

CSE-214 Online - 2 (C2)

Structural Design Pattern

Duration: 20 minutes

A retail company operates a customer loyalty programme with three membership tiers: **Regular**, **Premium**, and **Gold**. Customers earn loyalty points on every purchase. The number of points earned depends on the customer's membership tier:

- **Regular:** 1 point for every £100 spent.
- **Premium:** 2 points for every £100 spent.
- **Gold:** 3 points for every £100 spent.

Customers can redeem their accumulated loyalty points using one of the following redemption methods:

- **Cashback:** Convert the points into an instant discount on the current purchase.
- **Coupon:** Convert the points into shopping coupons for future purchases.
- **Charity Donation:** Donate the monetary value of the redeemed points to the company's charity programme.

Any membership tier should support any redemption method. For example, a Gold member may choose cashback, while a Premium member may choose charity donation. The company also plans to introduce new membership tiers (e.g., Platinum, Diamond) and new redemption methods (e.g., gift cards, airline miles, partner vouchers) in the future. The system should be designed so that new membership tiers and new redemption methods can be added independently without requiring significant modifications to the existing implementation.